Lesson 3.19 - How to Calculate the Average Deviation

> **Video:** How To Calculate the Average Deviation by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Formula

The **average deviation** (signed, with **no** absolute value) is

$$A = \frac{\sum(x - \bar{x})}{n}$$

Compare it with the **mean absolute deviation** (Lesson 3.18), which uses $|x - \bar{x}|$.

Example

Data: 8, 12, 15, 18, 19, 23, 31

Mean: $\bar{x} = \frac{\sum x}{n} = \frac{126}{7} = 18$

Deviations:

x	8	12	15	18	19	23	31
$x - 18$	−10	−6	−3	0	1	5	13

$$A = \frac{\sum(x - \bar{x})}{n} = \frac{-10 - 6 - 3 + 0 + 1 + 5 + 13}{7} = \frac{0}{7} = 0$$

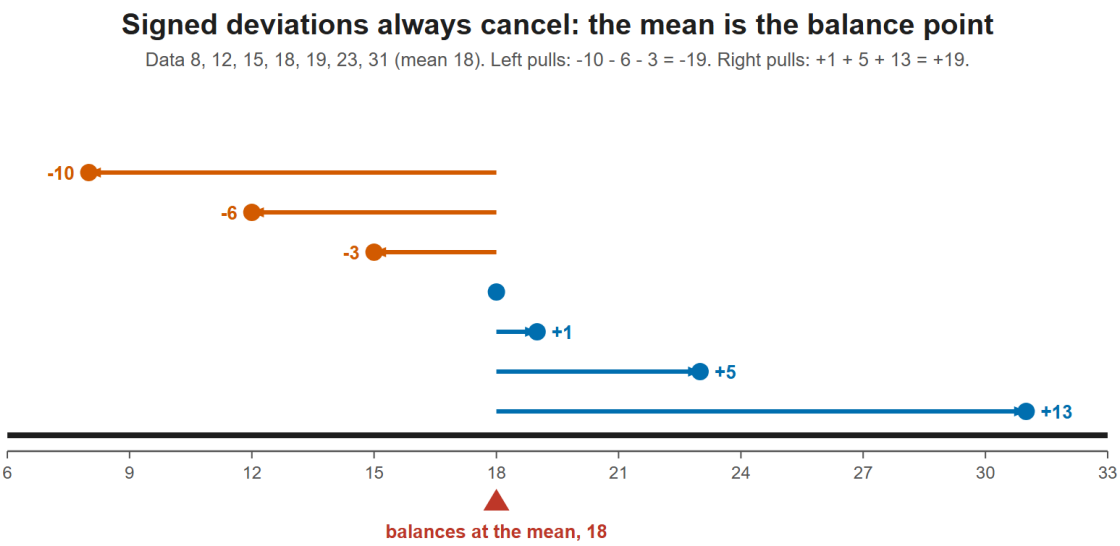
The positives and negatives cancel: -6 with $1 + 5$, and $-10 - 3 = -13$ with $+13$.

Why is it always zero?

$$\sum(x - \bar{x}) = \sum x - n\bar{x} = \sum x - \sum x = 0$$

This holds for **every** data set: the mean is exactly the balance point of the data. So the signed average deviation **cannot** measure spread, which is why statistics uses:

- the **mean absolute deviation**, which uses absolute values, and
- the **variance / standard deviation**, which square the deviations.



The leftward pulls (-10, -6, -3) add to -19 and the rightward pulls (+1, +5, +13) add to +19, so the signed deviations always total zero. The mean is the point where the data balance.

Key takeaways

- Average (signed) deviation = $\sum(x - \bar{x})/n$, and it is **always 0**.
- It is a useful check: if your deviations don't add to 0, the mean is wrong.
- To measure spread, use the MAD or the standard deviation instead.